# Quick Reference:



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Enterprise Architecture with TOGAF® and ArchiMate®

# The TOGAF® 9.1 Components

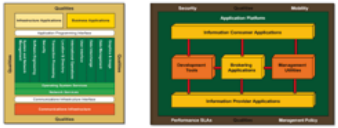
## ADM




## Architecture Content Framework



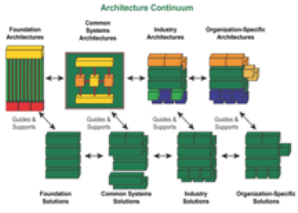
## Reference Models



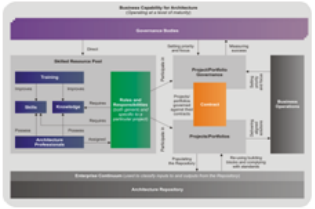
## ADM Guidelines & Techniques



## Enterprise Continuum



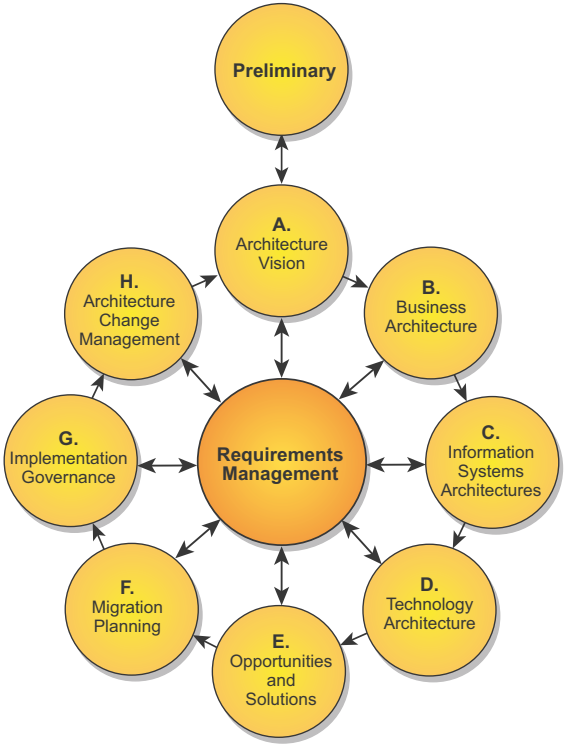

## Architecture Capability Framework



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* From TOGAF Version 9.1, The Open Group, 2011

# The TOGAF Architecture Development Method (ADM)




* From TOGAF Version 9.1, The Open Group, 2011

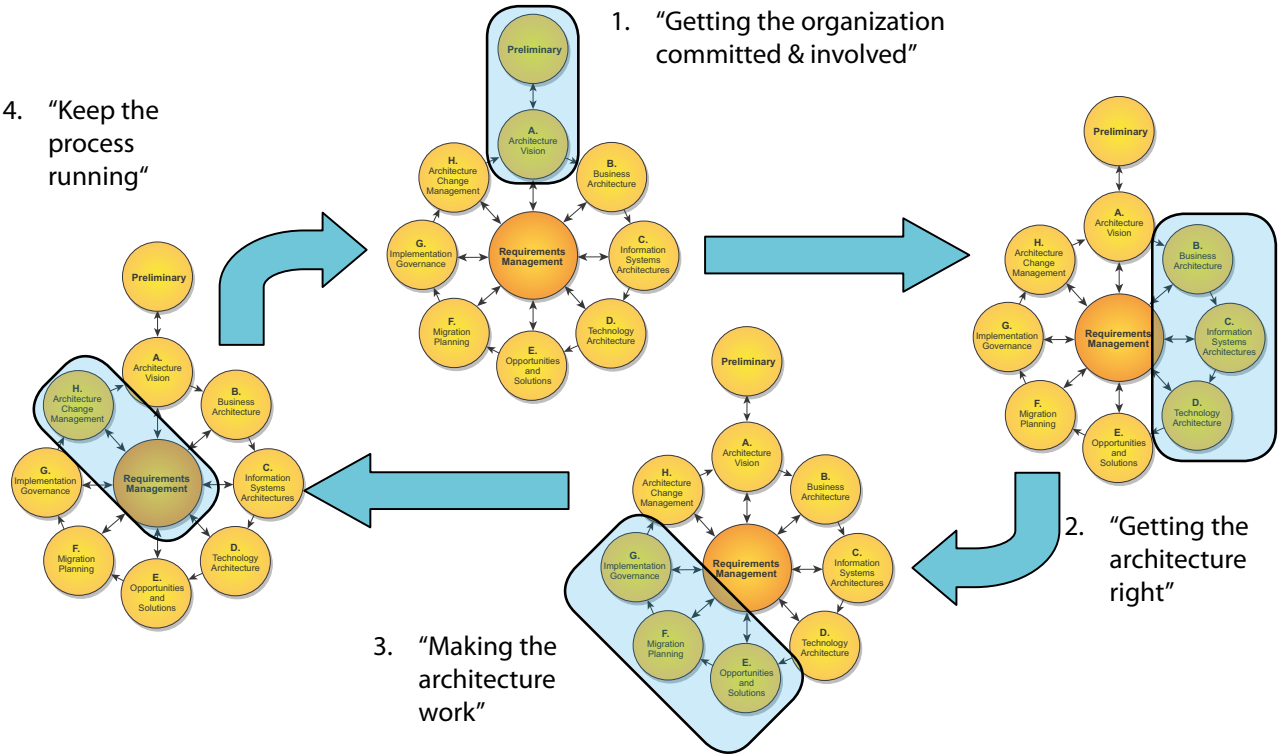
# Summary of the ADM process

1. "Getting the organization committed & involved"

2. "Getting the architecture right"

3. "Making the architecture work"

4. "Keep the process running"





**Input**

1. Reference Materials External: TOGAF, other EA frameworks
2. Non-Architectural Inputs: business strategies, principles, goals, and drivers, portfolio/project management, (architecture) governance, IT strategy
3. Architectural Inputs: Pre-existing models can be used as a base-line (e.g., Existing Organizational Model for EA, EA Framework, EA Principles, EA Repository)

**Objectives**

1. Determine the Architecture Capability desired by the organization
2. Establish the Architecture Capability

**Preliminary Phase**

**Steps**

1. Scope the Enterprise Organizations Impacted
2. Confirm Governance and Support Frameworks
3. Define and Establish Enterprise Architecture Team and Organization
4. Identify and Establish Architecture Principles
5. Select and Tailor Architecture Frameworks
6. Implement Architecture Tools

**Output**

1. Organizational Model for EA
2. Tailored Architecture Framework
3. Configured and deployed tools
4. Initial EA Repository
5. Request for EA Work
6. Governance Framework



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Business Principles, Business Goals, and Business Drivers
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework

**Objectives**

1. Develop a high-level aspirational vision of the capabilities and business value to be delivered as a result of the proposed enterprise architecture
2. Obtain approval for a Statement of Architecture Work that defines a program of works to develop and deploy the architecture outlined in the Architecture Vision

**Steps**

1. Establish the Architecture Project
2. Identify Stakeholders, Concerns, and Business Requirements
3. Confirm and Elaborate Business Goals, Business Drivers, and Constraints
4. Evaluate Business Capabilities
5. Assess Readiness for Business Transformation
6. Define Scope
7. Confirm and Elaborate Architecture Principles
8. Develop Architecture Vision
9. Define the Target Architecture Value Propositions and KPIs
10. Identify the Business Transformation Risks and Mitigation Activities
11. Develop Enterprise Architecture Plans and Statement of Architecture Work; Secure Approval

**Phase A: Architecture vision**

**Output**

1. Approved Statement ofArchitecture Work
2. Refined statements of Business Principles, Business Goals, and Business Drivers
3. Architecture Principles
4. Capability Assessment
5. Tailored Architecture Framework
6. Architecture Vision
7. Additional content populating the Architecture Repository



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Business Principles, Business Goals, and Business Drivers, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Approved Statement of Architecture Work, Architecture Principles, Enterprise Continuum, Architecture Repository, Architecture Vision

**Objectives**

1. Develop the Target Business Architecture that describes how the enterprise needs to operate to achieve the business goals, and respond to the strategic drivers set out in the Architecture Vision, in a way that addresses the Request for Architecture Work and stakeholder concerns
2. Identify candidate Architecture Roadmap components based upon gaps between the Baseline and Target Business Architectures

**Phase B: Business Architecture**

**Steps**

1. Select Reference Models, Viewpoints,and Tools
2. Develop Baseline BA Description
3. Develop Target BA Description
4. Perform Gap Analysis
5. Define Roadmap Components
6. Resolve Impacts Across the Architecture Landscape
7. Conduct Formal Stakeholder Review
8. Finalise the Business Architecture
9. Create Architecture Definition Document

**Output**

1. Refined and updated versions of the Architecture Vision phase deliverables
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. BA components of an Architecture Roadmap



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Application Principles, Data Principles, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document, Draft Architecture Requirements Specification, BA components of an Architecture Roadmap.

**Objectives**

1. Develop the Target Information Systems (Data and Application) Architecture, describing how the enterprise's Information Systems Architecture will enable the Business Architecture and the Architecture Vision, in a way that addresses the Request for Architecture Work and stakeholder concerns
2. Identify candidate Architecture Roadmap components based upon gaps between the Baseline and Target Information Systems (Data and Application) Architectures

## Phase C: Information Systems Architectures

**Output**

1. Refined and updated versions of the Architecture Vision phase Deliverables, including updated Statement of Architecture Work
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. Information systems components of an Architecture Roadmap

**Steps**

Detailed steps for Phase C are given separately for each architecture domain.



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Data Principles, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document, Draft Architecture Requirements Specification, BA components of an Architecture Roadmap.

**Objectives**

1. Develop the Target Data Architecture that enables the Business Architecture and the Architecture Vision, while addressing the Request for Architecture Work and stakeholder concerns
2. Identify candidate Architecture Roadmap components based upon gaps between the Baseline and Target Data Architectures.

## Phase C: Data Architecture

**Steps**

1. Select Reference Models, Viewpoints, and Tools
2. Develop Baseline Data Architecture Description
3. Develop Target Data Architecture Description
4. Perform Gap Analysis
5. Define Roadmap Components
6. Resolve Impacts Across the Architecture Landscape
7. Conduct Formal Stakeholder Review
8. Finalise the Data Architecture
9. Create Architecture Definition Document

**Output**

1. Refined and updated versions of the Architecture Vision phase Deliverables, including updated Statement of Architecture Work
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. DA components of an Architecture Roadmap



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Application Principles, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document, Draft Architecture Requirements Specification, BA and DA components of an Architecture roadmap.

**Objectives**

1. Develop the Target Application Architecture that enables the Business Architecture and the Architecture Vision, while addressing the Request for Architecture Work and stakeholder concerns
2. Identify candidate Architecture Roadmap components based

## Phase C: Application Architecture

**Steps**

1. Select Reference Models, Viewpoints, and Tools
2. Develop Baseline Application Architecture Description
3. Develop Target Application Architecture Description
4. Perform Gap Analysis
5. Define Roadmap Components
6. Resolve Impacts Across the Architecture Landscape
7. Conduct Formal Stakeholder Review
8. Finalise the Application Architecture
9. Create Architecture Definition Document

**Output**

1. Refined and updated versions of the Architecture Vision phase Deliverables, including updated Statement of Architecture Work, Validated or new application principles
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. AA components of an Architecture Roadmap



**Input**

1. Reference Materials External: Architecture reference materials, Product information on candidate products
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Technology Principles, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document, Draft Architecture Requirements Specification, BA, DA and AA components of an Architecture Roadmap.

**Objectives**

1. Develop the Target Technology Architecture that enables the logical and physical application and data components and the Architecture Vision, addressing the Request for Architecture Work and stakeholder concerns
2. Identify candidate Architecture Roadmap components based upon gaps between the Baseline and Target Technology Architectures

## Phase D: Technology Architecture

**Steps**

1. Select Reference Models, Viewpoints, and Tools
2. Develop Baseline Application Architecture Description
3. Develop Target Application Architecture Description
4. Perform Gap Analysis
5. Define Roadmap Components
6. Resolve Impacts Across the Architecture Landscape
7. Conduct Formal Stakeholder Review
8. Finalise the Application Architecture
9. Create Architecture Definition Document

**Output**

1. Refined and updated versions of the Architecture Vision phase Deliverables, including updated Statement of Architecture Work, Validated or new Technology principles
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. TA components of an Architecture Roadmap



**Input**

1. Reference Materials External: Architecture referencematerials, Product information
2. Non-Architectural Inputs:Request for ArchitectureWork, Capability Assessment, Communication Plan, Planning Methodologies
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Governance models and frameworks, Tailored Architecture Framework, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document,Draft ArchitectureRequirementsSpecification,Change requests.

**Objectives**

1. Generate the initial complete version of the Architecture Roadmap, based upon the gap analysis and candidate Architecture Roadmap components from Phases B, C, and D
2. Determine whether an incremental approach is required, and if so identify Transition Architectures that will deliver continuous business value

## Phase E: Opportunities And Solutions

**1. Steps**

2. Determine/Confirm Key Corporate Change Attributes
3. Determine Business Constraints for Implementation
4. Review/Consolidate Gap Analysis Results
5. Review IT Requirements from a Functional Perspective
6. Consolidate/Reconcile Interoperability Requirements
7. Refine and Validate Dependencies
8. Confirm Readiness/Risk for Business Transformation
9. Formulate High-Level Implementation/Migration Strategy
10. Identify and Group Major Work Packages
11. Identify Transition Architectures
12. Create Portfolio/Project Charters, Update Architectures

**Output**

1. Refined and updated versions of the Architecture Vision, BA, ISA and TA phases Deliverables, including Updated Statement of Architecture Work, Architecture Vision
2. Draft Architecture Definition Document
3. Draft Architecture Requirements Specification
4. Consolidated and validated Architecture Roadmap
5. Capability Assessment,
6. Transition Architecture, Version 1.0
7. Implementation and Migration Plan, Version 0.1, including the high-level Implementation and Migration Strategy



**Input**

1. Reference Materials External:Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment, Communication Plan
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Governance models and frameworks, Statement of Architecture Work, Architecture Vision, Architecture Repository, Draft Architecture Definition Document, Draft Architecture Require-ments Specification, Change requests, Consolidated Architecture Roadmap, Capability Assessment, Transition Architecture V 1.0, Implementation and Migration Plan V 1.0.

**Steps**

1. Confirm Management Framework Interactions for Implementation and Migration Plan
2. Assign a Business Value to Each Project
3. Estimate Resource Requirements, Project Timings, and Availability/Delivery Vehicles
4. Prioritise Migration Projects through Cost/Benefit Assessment/Risk Validation
5. Confirm Transition Architecture Increments/Phases and Update Architecture Definition Document
6. Generate the Architecture Implementation Roadmap and Migration Plan
7. Establish the Architecture Evolution Cycle and Document Lessons Learned

**Phase F: Migration Planning**

**Objectives**

1. Finalize the Architecture Roadmap and the supporting Implementation and Migration Plan
2. Ensure that the Implementation and Migration Plan is coordinated with the enterprise's approach to managing and implementing change in the enterprise's overall change portfolio
3. Ensure that the business value and cost of work packages and Transition Architectures is understood by key stakeholders

**Output**

1. Implementation & Migration Plan,V1.0
2. Finalised Architecture Definition Document
3. Finalised Architecture Requirements Spec.
4. Finalised Architecture Roadmap
5. Finalised Transition Architecture
6. Re-Usable Architecture Building Blocks
7. Requests for Architecture Work for the architecture aspects of implementation projects
8. Architecture Contracts for implementation projects
9. Implementation Governance Model
10. Change Requests from lessons learned



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work, Capability Assessment
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Statement of Architecture Work, Architecture Vision, Architecture Repository, Architecture Definition Document, Architecture Requirements Specification, Architecture Roadmap, Transition Architecture including Implementation Governance Model, Architecture Contract, Request for Architecture Work (from phase E and F), Implementation and Migration Plan

**Objectives**

1. Ensure conformance with the Target Architecture by implementation projects
2. Perform appropriate Architecture Governance functions for the solution and any implementation-driven architecture Change Requests

## Phase G: Implementation Governance

**Steps**

1. Confirm Scope and Priorities for Deployment with Development Management
2. Identify Deployment Resources and Skills
3. Guide Development of Solutions Deployment
4. Perform Enterprise Architecture Compliance Reviews
5. Implement Business and IT Operations
6. Perform Post-Implementation Review and Close Implementation

**Output**

1. Architecture Contract (signed), as recommended in the architecture-compliant implemented architectures
2. Compliance Assessments
3. Change Requests
4. Architecture-compliant solutions deployed



**Input**

1. Reference Materials External: Architecture reference materials
2. Non-Architectural Inputs: Request for Architecture Work from phases E & F
3. Architectural Inputs: Organizational Model for Enterprise Architecture, Tailored Architecture Framework, Statement of Architecture Work, Architecture Vision, Architecture Repository, Architecture Definition Document, Architecture Requirements Specification, Architecture Roadmap, Change Request – technology & business changes, from lessons learned, Transition Architecture including Implementation Governance Model, Architecture Contract, Request for Compliance Assessments, Implementation and Migration Plan

**Objectives**

1. Ensure that the architecture lifecycle is maintained
2. Ensure that the Architecture Governance Framework is executed
3. Ensure that the enterprise Architecture Capability meets current requirements

**Phase H: Architecture Change Management**

**Steps**

1. Establish Value Realisation Process
2. Deploy Monitoring Tools
3. Manage Risks
4. Provide Analysis for Architecture Change Management
5. Develop Change Requirements to Meet Performance Targets
6. Manage Governance Process
7. Activate the Process to Implement Change

**Output**

1. Architecture Contract (signed), as Requirements Impact Assessment
2. Architecture Requirements Specification



**Input**
1. Architecture Repository
2. Organizational Model for EA
3. Tailored Architecture Framework,
4. Statement of Architecture Work
5. Architecture Vision
6. Architecture Requirements
7. Requirements Impact Assessment

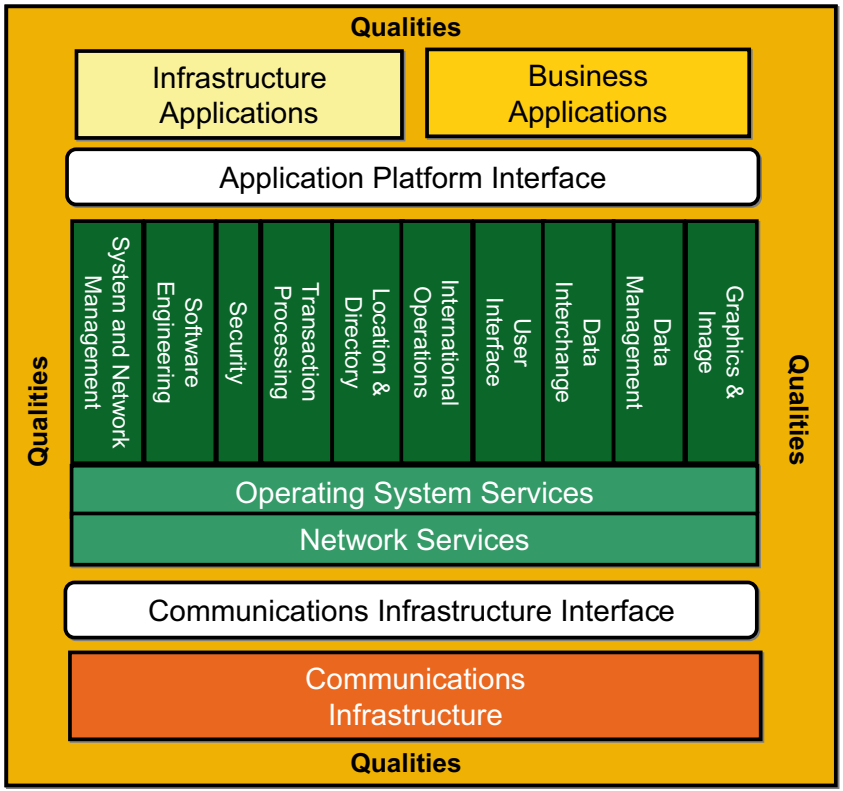
**Objectives**
1. Ensure that the Requirements Management process is sustained and operates for all relevant ADM phases
2. Manage architecture requirements identified during any execution of the ADM cycle or a phase
3. Ensure that relevant architecture requirements are available for use by each phase as the phase is executed

**Steps**
1. Identify/document requirements (ADM phase)
2. Baseline requirements (RM)
3. Monitor baseline requirements (RM)
4. Identify changed requirements (ADM phase)
5. Identify changed requirements and record priorities (RM)
6. Assess impact of changed requirements on current phase, assess impact of changed requirements on previous phases, determine whether to implement change, or defer to later ADM cycle, issue Requirements Impact Statement, V n+1 (ADM phase)
7. Implement requirements arising from Phase H (ADM phase)
8. Update the requirements repository with information relating to the changes requested, including stakeholder views affected (RM)
9. Implement change in the current phase (ADM phase)
10. Assess and revise gap analysis for past phases (ADM phase)



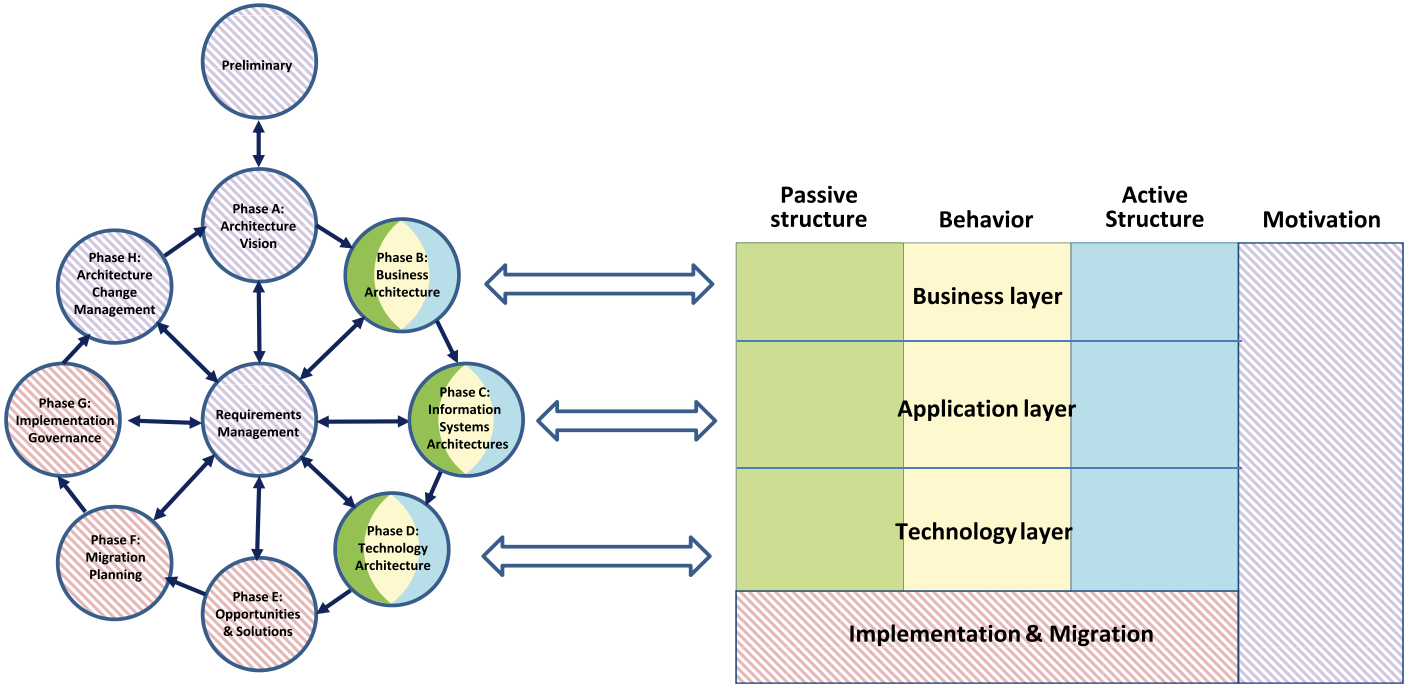

**Output**
1. Requirements Impact Assessment
2. Architecture Requirements Specification

# TOGAF Technical Reference Model (TRM)




* From TOGAF Version 9.1, The Open Group, 2011

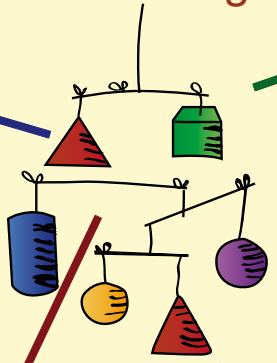
# ArchiMate® framework

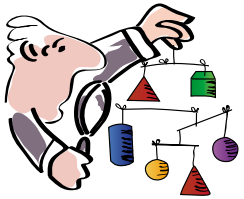



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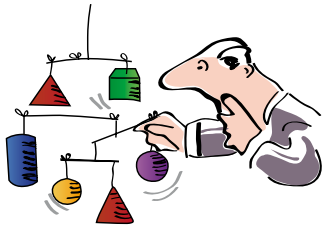
High-level modeling *within* a domain

ArchiMate language





Basis for *visualizations*

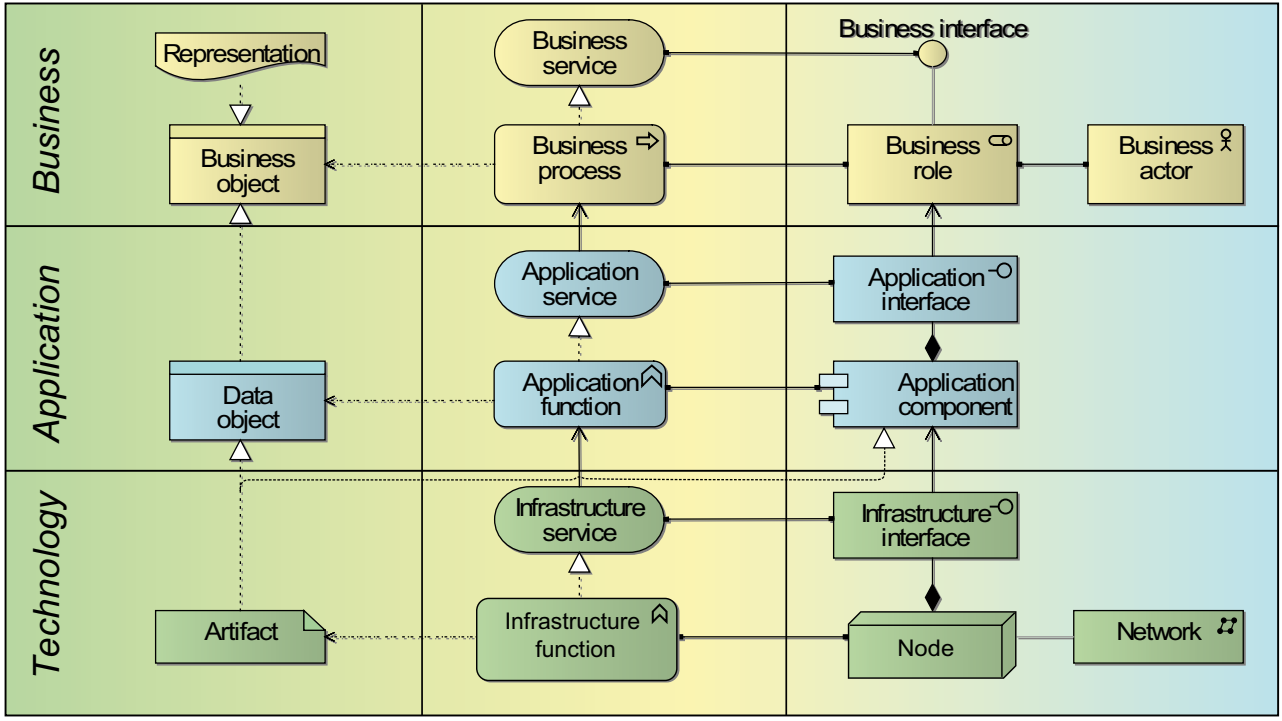


Basis for *analysis*

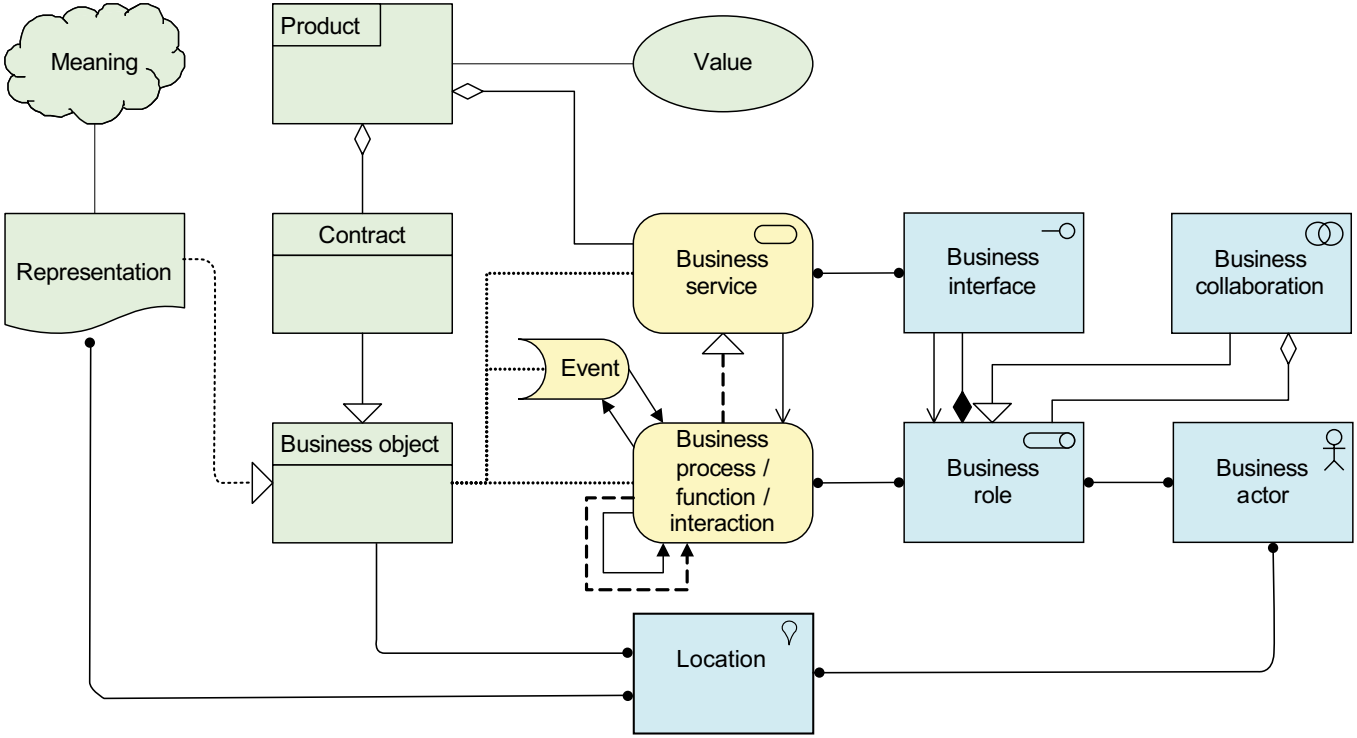
Modeling relations *between* domains

Relate to *standards*

# ArchiMate language summary



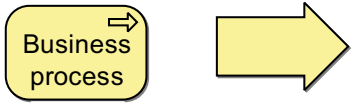
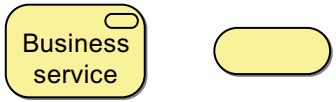
# ArchiMate business layer metamodel



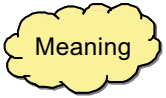
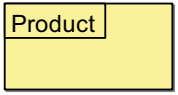
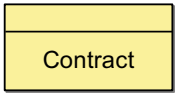
# ArchiMate business layer concepts (1)

| Concept | Definition |
|---|---|
| Business actor  | An organizational entity that is capable of performing behavior |
| Business role  | The responsibility for performing specific behavior, to which an actor can be assigned |
| Business collaboration  | An aggregate of two or more business roles that work together to perform collective behavior |
| Business interface  | A point of access where a business service is made available to the environment |
| Location  | A conceptual point or extent in space |
| Business object | A passive element that has relevance from a business perspective |

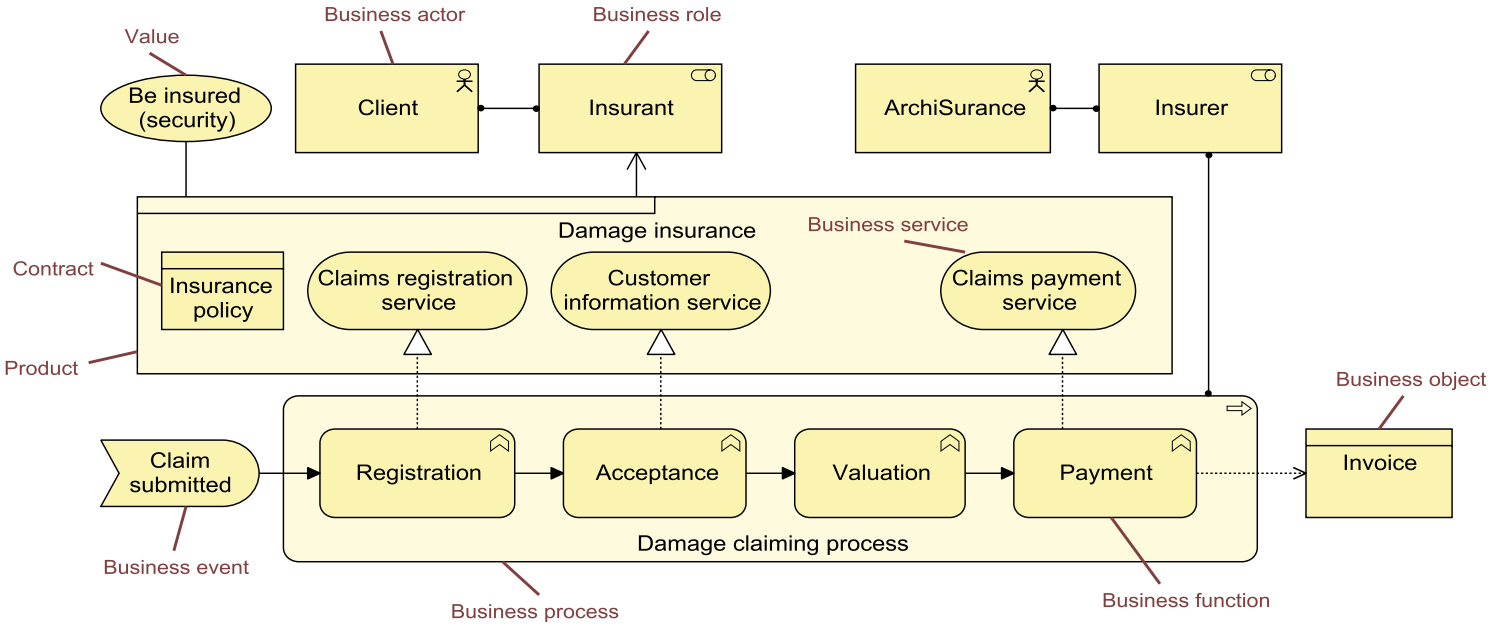
# ArchiMate business layer concepts (2)

| Concept | Definition |
|---|---|
|   | A behavior element that groups behavior based on an ordering of activities; it is intended to produce a defined set of products or business services |
|   | A behavior element that groups behavior based on a chosen set of criteria (typically required business resources and/or competences). |
|   | A behavior element that describes the behavior of a business collaboration |
|   | Something that happens (internally or externally) and influences behavior |
|   | A service that fulfills a business need for a customer (internal or external to the organization). |

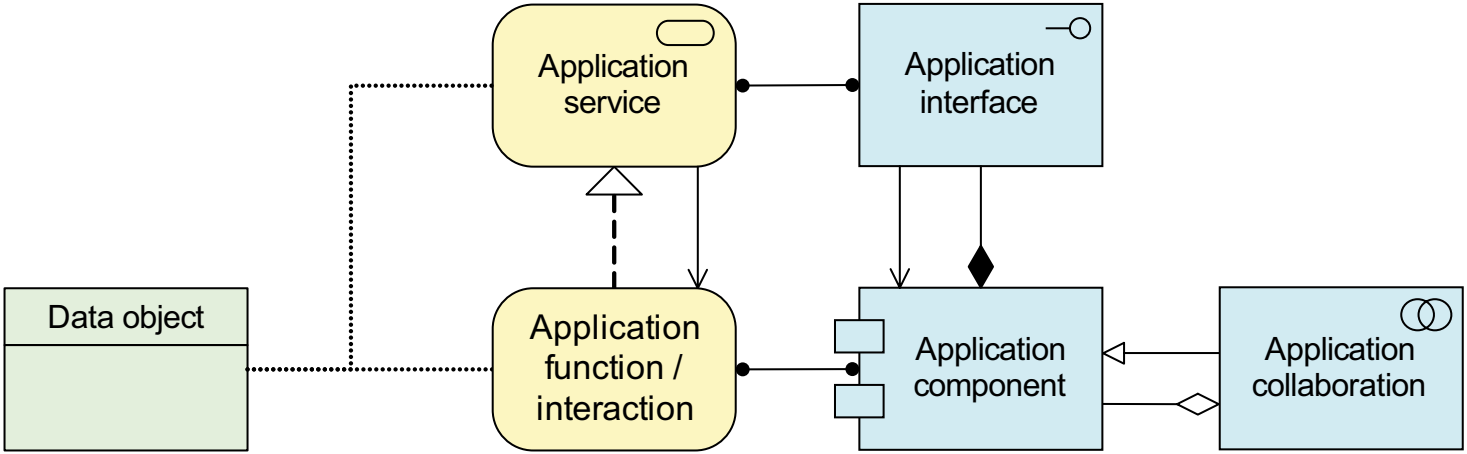
# ArchiMate business layer concepts (3)

| Concept | Definition |
|---|---|
|  | The perceptible form of the information carried by a business object |
|  | The knowledge or expertise present in the representation of a business object, given a particular context |
|  | The relative worth, utility, or importance of a business service or product |
|  | A coherent collection of services, accompanied by a contract/set of agreements, which is offered as a whole to (internal or external) customers |
|  | A formal or informal specification of agreement that specifies the rights and obligations associated with a product |

# ArchiMate business layer example model



# ArchiMate application layer metamodel



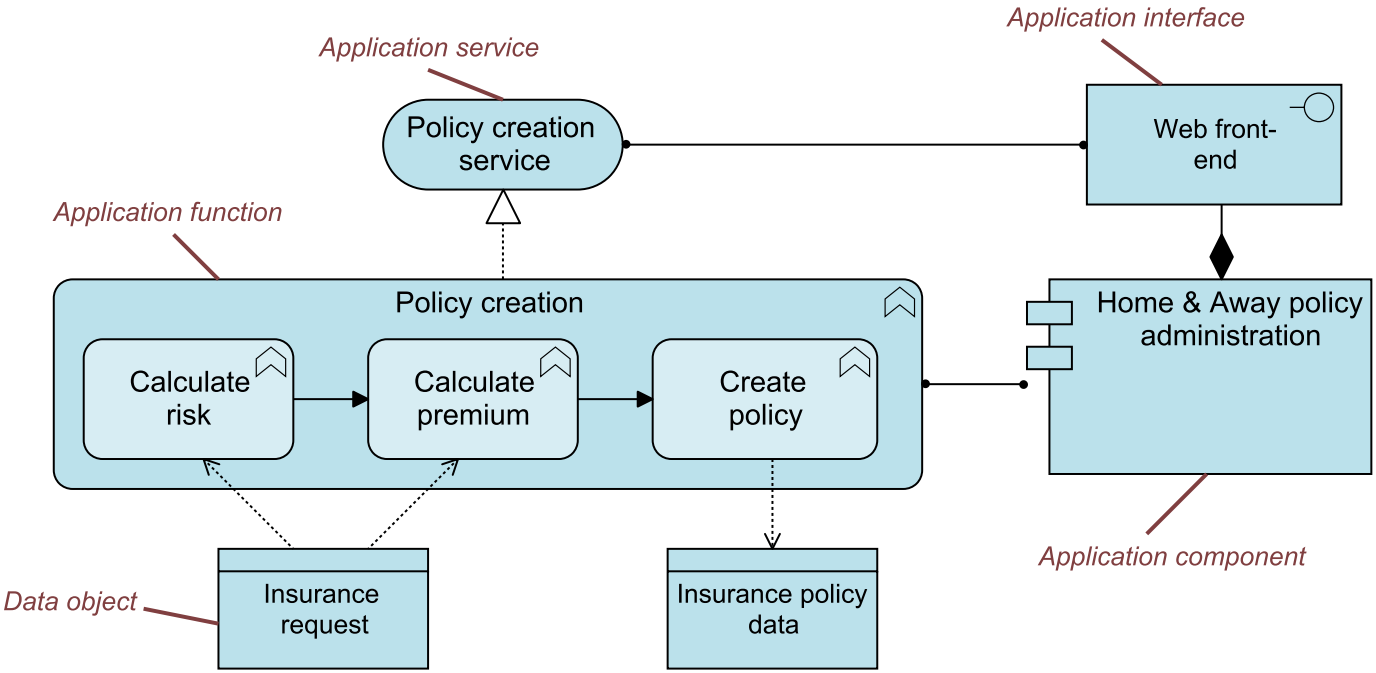
# ArchiMate application layer concepts (1)

| Concept | Definition |
|---|---|
| Application component  | A modular, deployable, and replaceable part of a system that encapsulates its contents and exposes its functionality through a set of interfaces |
| Application collaboration  | An aggregate of two or more application components that work together to perform collective behavior |
| Application interface  | A point of access where an application service is made available to a user or another application component |
| Data object | A passive element suitable for automated processing |

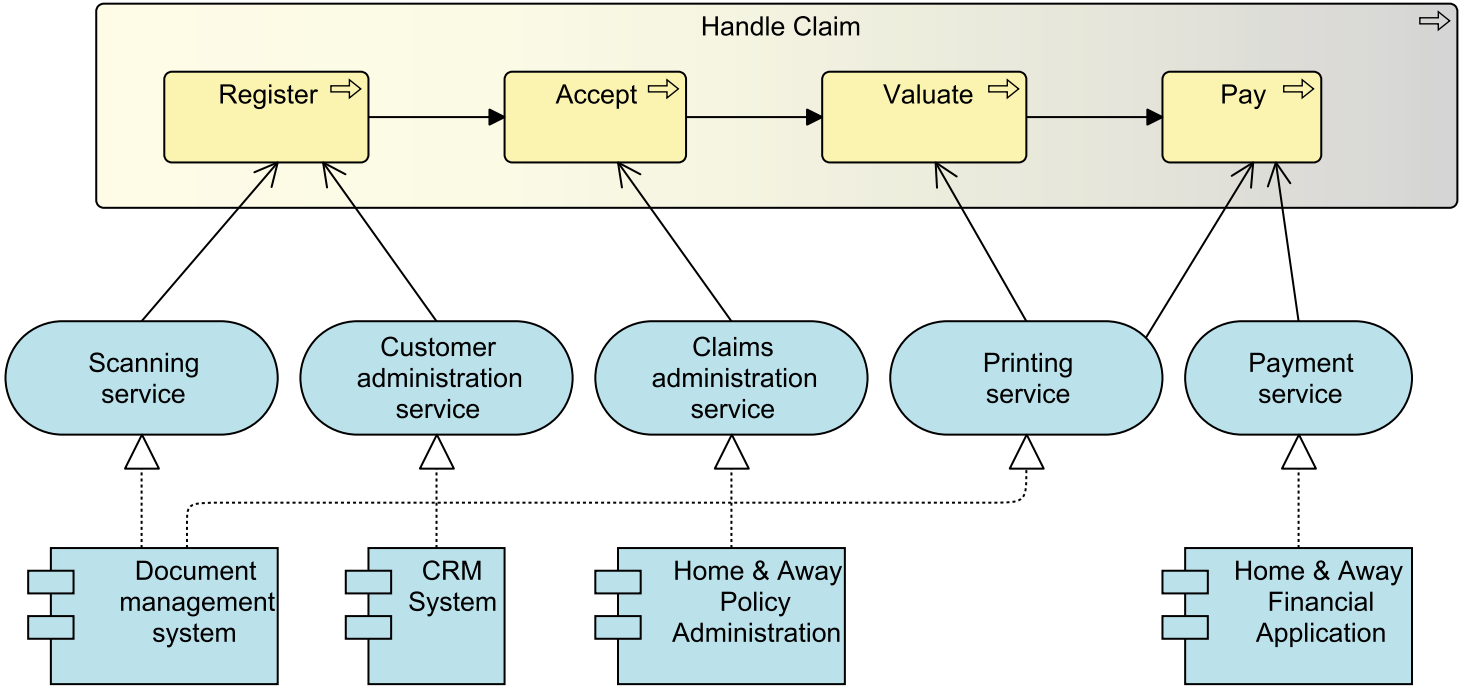
# ArchiMate application layer concepts (2)

| Concept | Definition |
| --- | --- |
| Application function  | A behavior element that groups automated behavior that can be performed by an application component |
| Application interaction  | A behavior element that describes the behavior of an application collaboration |
| Application service | A service that exposes automated behavior |

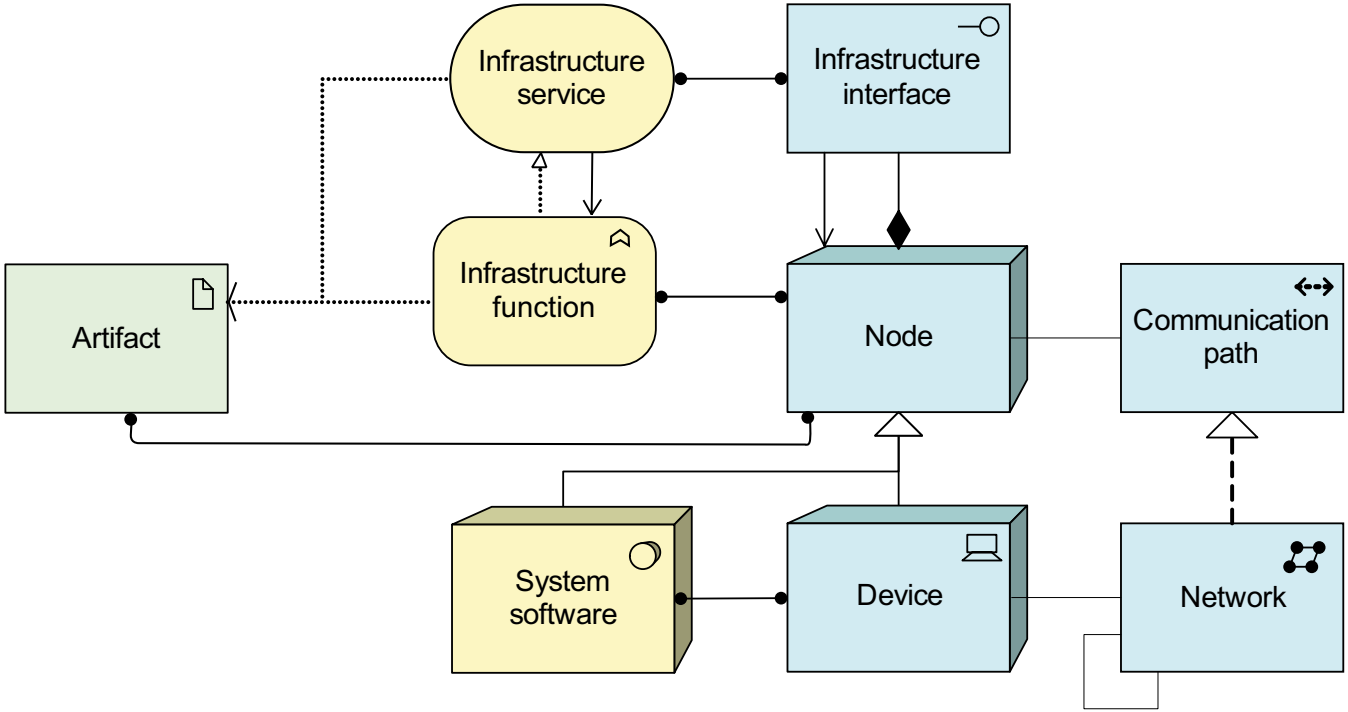
# ArchiMate application layer example model



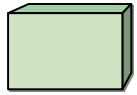
# Process/application support example



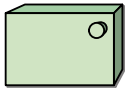
# ArchiMate technology layer metamodel



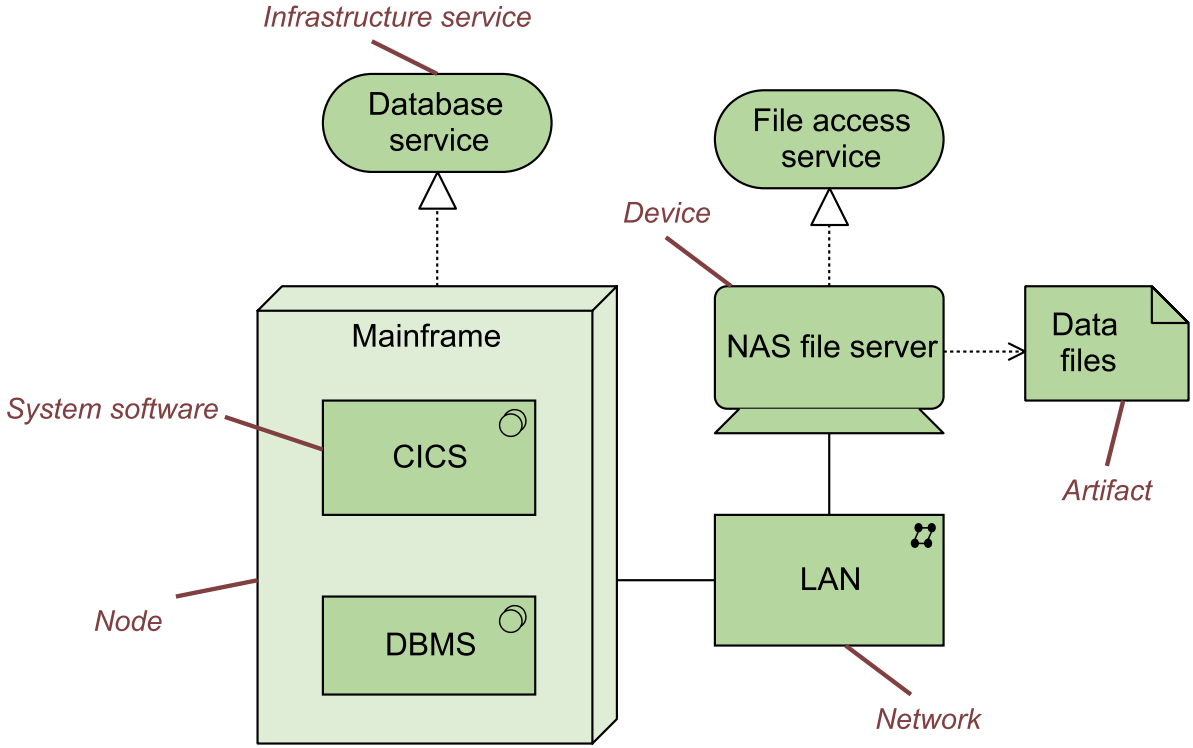
# ArchiMate technology layer concepts (1)

| Concept | Definition |
| --- | --- |
| Node  | A computational resource upon which artifacts may be deployed for execution |
| Device  | A hardware resource upon which artifacts may be stored or deployed for execution |
| Network  | A communication medium between two or more devices |
| Communication path  | A link between two or more nodes, through which these nodes can exchange data |

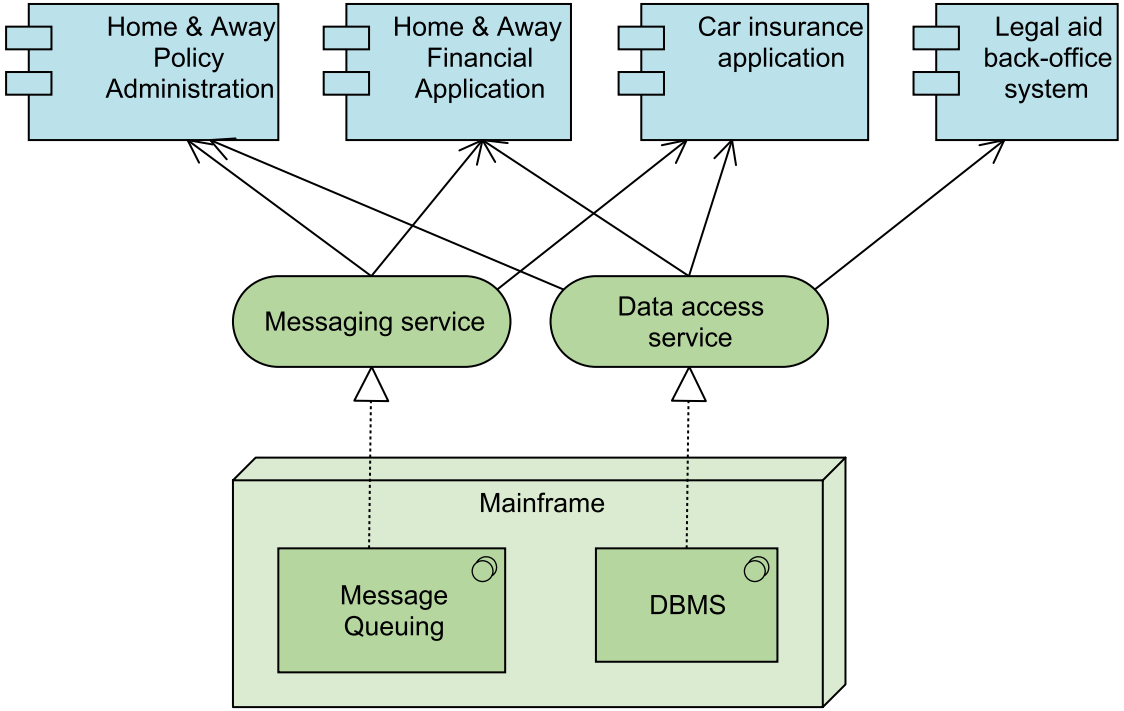
# ArchiMate technology layer concepts (2)

| Concept | Definition |
|---|---|
| Infrastructure interface  | A point of access where the functionality offered by a node can be accessed by other nodes and application components |
| System software  | A software environment for specific types of components and objects that are deployed on it in the form of artifacts |
| Infrastructure function  | A behavior element that groups infrastructural behavior that can be performed by a node |
| Infrastructure service | An externally visible unit of functionality, provided by one or more nodes, exposed through well-defined interfaces, and meaningful to the environment |
| Artifact  | A physical piece of information that is used or produced in a software development process, or by deployment and operation of a system |

# ArchiMate technology layer example model



# Application-technology support example



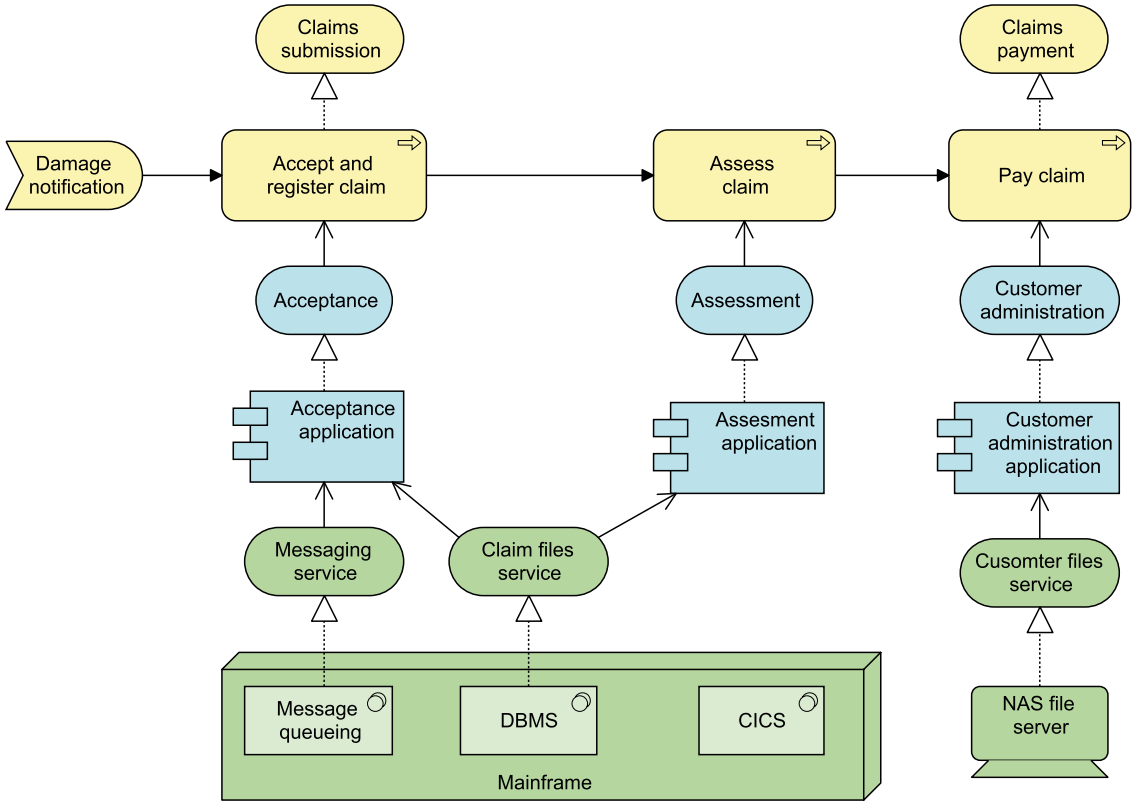
# ArchiMate structural relationships

| Relationship | Definition |
|---|---|
| Association  | A relationship between objects that is not covered by another, more specific relationship |
| Access  | The access of behavioral concepts to business or data objects |
| Used by  | The use of services by processes, functions, or interactions and the access to interfaces by roles, components, or collaborations |
| Realization  | Links a logical entity with a more concrete entity that realizes it |
| Assignment  | Links units of behavior with active elements (e.g., roles, components) that perform them, or roles with actors that fulfill them |
| Aggregation  | Indicates that an object groups a number of other objects |
| Composition  | Indicates that an object consists of a number of other objects |

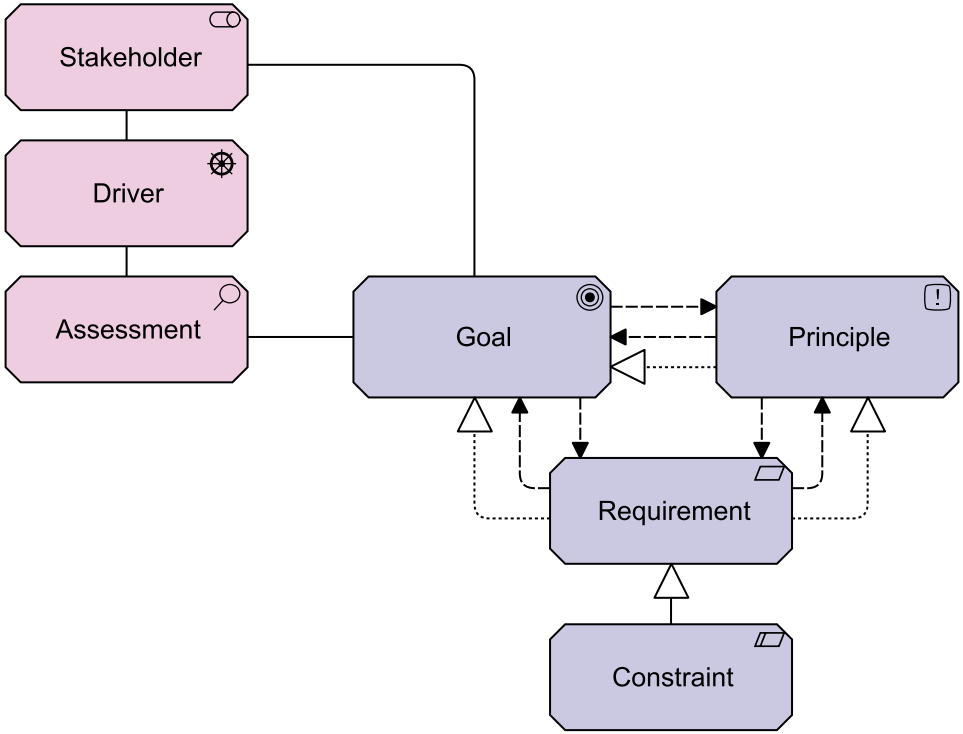
# ArchiMate dynamic and other relationships

| Relationship | Definition |
|---|---|
| Flow  | Describes the exchange or transfer of, for example, information or value between processes, function, interactions, and events |
| Triggering  | Describes the temporal or causal relations between processes, functions, interactions, and events |
| Grouping  | Indicates that objects, of the same type or different types, belong together based on some common characteristic |
| Junction  | Is used to connect relationships of the same type |
| Specialization  | Indicates that an object is a specialization of another object |

# ArchiMate layered view example



# ArchiMate motivation extension metamodel



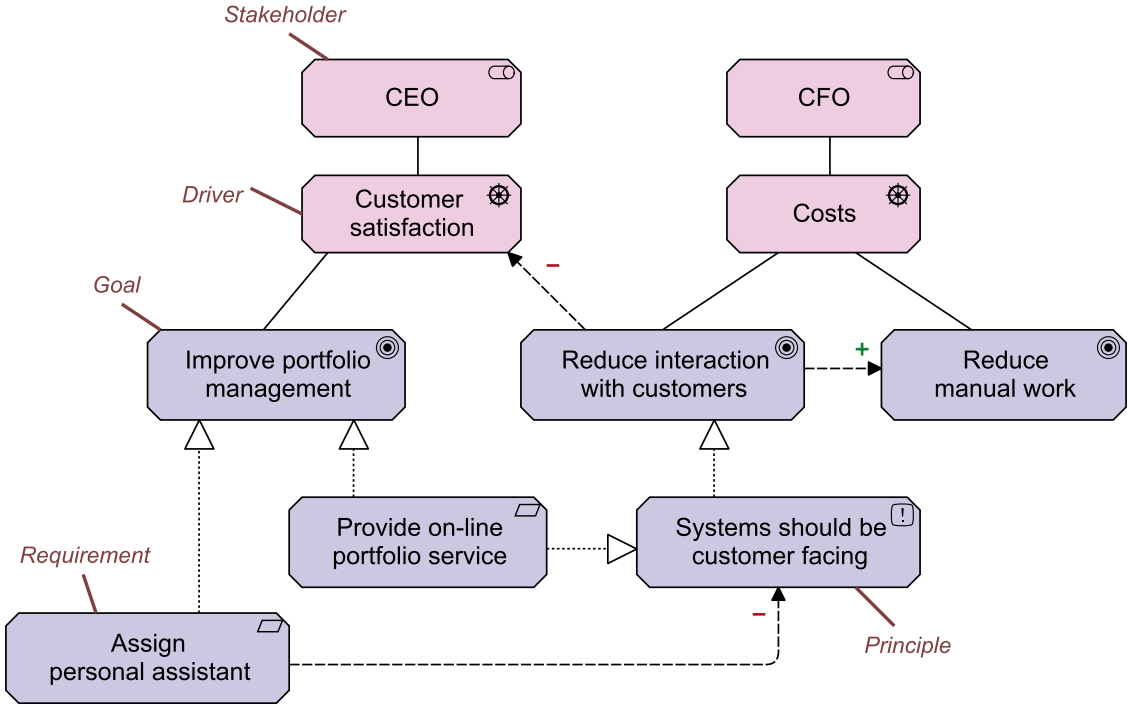
# ArchiMate motivation extension concepts

| Concept | Definition |
|---|---|
| Stakeholder | The role of an individual, team, or organization (or classes thereof) that represents their interests in, or concerns relative to, the outcome of the architecture |
| Driver | Something that creates, motivates, and fuels the change in an organization |
| Assessment | The outcome of some analysis of some driver |
| Goal | An end state that a stakeholder intends to achieve |
| Principle | A normative property of all systems in a given context, or the way in which they are realized. |
| Requirement  | A statement of need that must be realized by a system |
| Constraint  | A restriction on the way in which a system is realized |

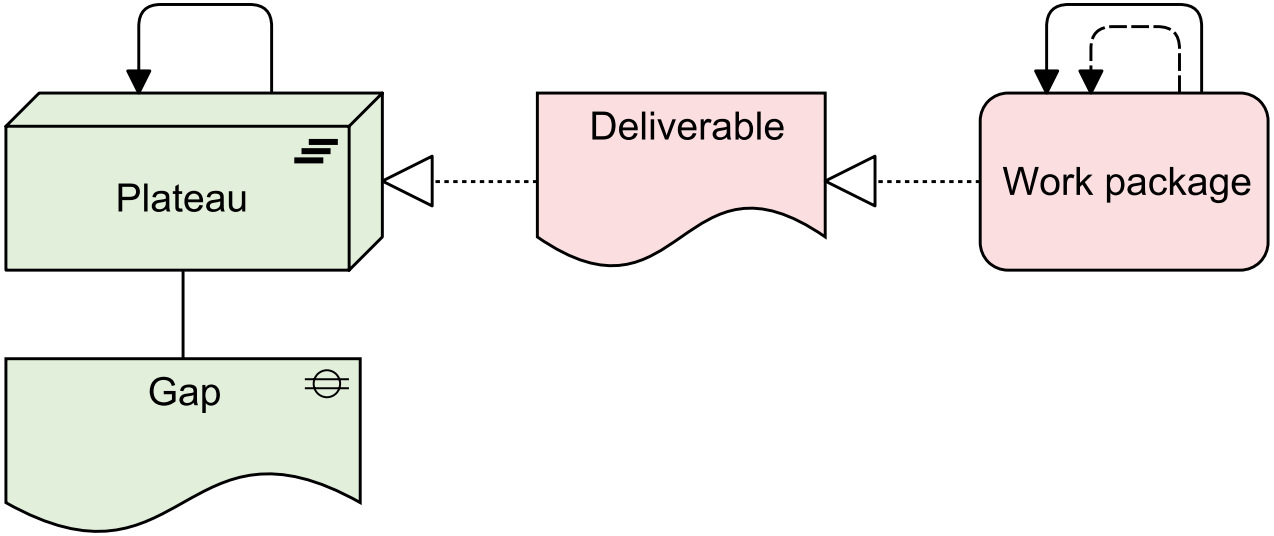
# ArchiMate motivation extension relationships

| Relationship | Definition |
|---|---|
| Aggregation (decomposition)  | Models that some intentional element is divided into multiple intentional elements |
| Realization (means-end)  | Models that some end is realized by some means |
| Specialization  | Indicates that an object is a specialization of another object (of the same type) |
| Association  | Is re-used from the ArchiMate Core to relate stakeholders to concerns, concerns to assessment, and assessments to goals |
| Influence   | Models that some motivational element has a positive or negative influence on the realization of another motivational element |

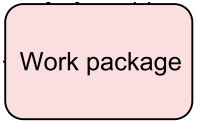
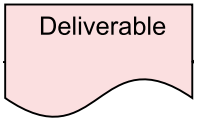
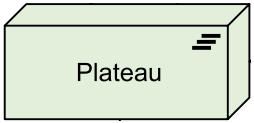
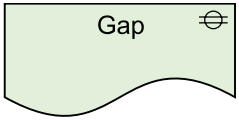
# ArchiMate motivation extension metamodel



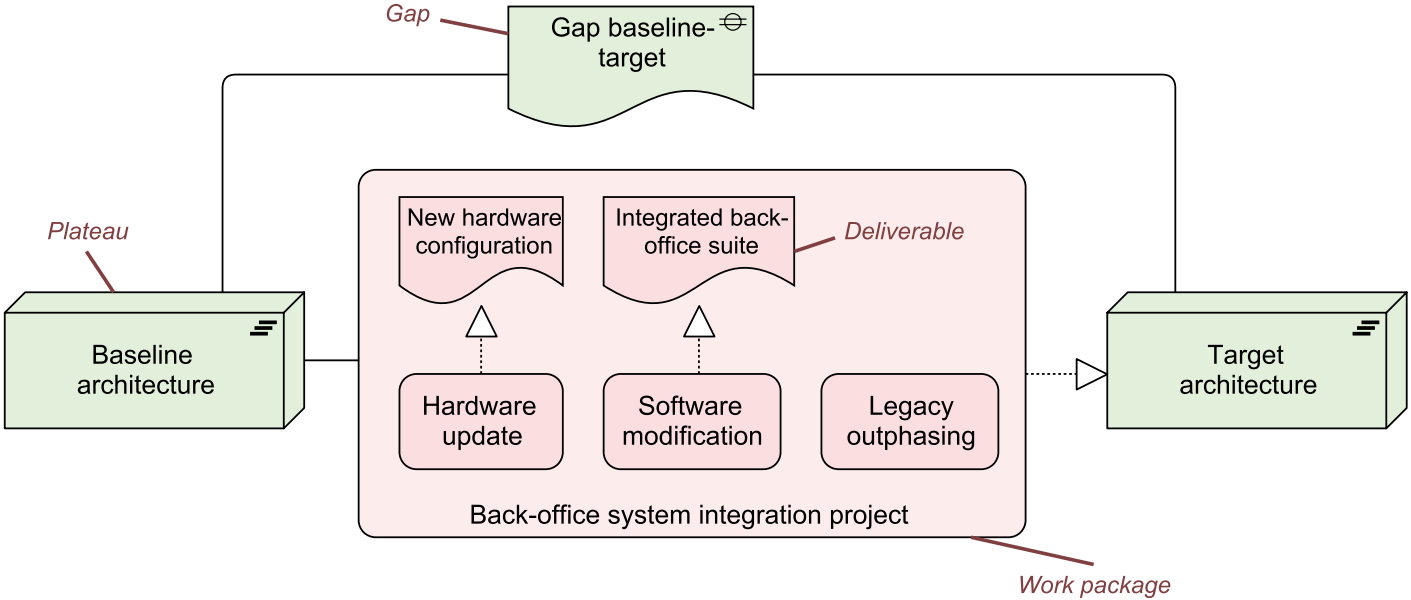
# ArchiMate implementation & migration extension metamodel



# ArchiMate implementation & migration extension concepts

| Concept | Definition |
|---|---|
|  | A series of actions designed to accomplish a unique goal within a specified time |
|  | A precisely defined outcome of a work package |
|  | A relatively stable state of the architecture that exists during a limited period of time |
|  | An outcome of a gap analysis between two plateaus |

# ArchiMate implementation & migration extension example model



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